

Sample Components and Values Used in Preparation

Chemical entry	Recorded names	Rows / systems	Value used	Chemical states used	Treatment in the prepared system	Source of value
Buffers						
Acetate	acetate; sodium acetate; ammonium acetate; NaAc; deuterated sodium acetate; Acetate buffer Na	17 / 17	$pK_a = 4.76$	$\text{CH}_3\text{COOH}(0) \rightleftharpoons \text{CH}_3\text{COO}^-(-1)$	Not included as a molecule; calculated contribution added to the target ionic strength.	(Goldberg et al. 2002)
Bis-Tris	Bis-Tris; Bis-Tris buffer	3 / 3	$pK_a = 6.46$	$\text{Bis-Tris-H}^+(+1) \rightleftharpoons \text{Bis-Tris}(0)$	Not included as a molecule; calculated contribution added to the target ionic strength.	(Goldberg et al. 2002)
Citrate	ammonium citrate	1 / 1	$pK_a = 3.13, 4.76, 6.40$	$\text{H}_3\text{Cit}(0) \rightleftharpoons \text{H}_2\text{Cit}^-(-1) \rightleftharpoons \text{HCit}^{2-}(-2) \rightleftharpoons \text{Cit}^{3-}(-3)$	Not included as a molecule; calculated contribution added to the target ionic strength.	(Goldberg et al. 2002)
HEPES	Hepes	1 / 1	$pK_a = 7.55$	$\text{HEPES-H}^{\pm}(0) \rightleftharpoons \text{HEPES}^-(-1)$	Not included as a molecule; calculated contribution added to the target ionic strength.	(Goldberg et al. 2002)
MES	MES	9 / 9	$pK_a = 6.10$	Executed: $\text{MES-H}^+(+1) \rightleftharpoons \text{MES}(0)$. Source: $\text{MES-H}^{\pm}(0) \rightleftharpoons \text{MES}^-(-1)$	Not included as a molecule; executed charge states used for the target ionic strength.	(Goldberg et al. 2002)
Phosphate	phosphate; sodium/potassium phosphate; phosphate buffer; NaPi/NaPI; KPi/KPO4; Na2PO4; named mono-/dibasic mixtures	82 / 82	$pK_a = 2.15, 7.21, 12.40$	$\text{H}_3\text{PO}_4(0) \rightleftharpoons \text{H}_2\text{PO}_4^-(-1) \rightleftharpoons \text{HPO}_4^{2-}(-2) \rightleftharpoons \text{PO}_4^{3-}(-3)$	Not included as a molecule; calculated contribution added to the target ionic strength.	(Goldberg et al. 2002)
Tris	Tris; Tris-HCl; d-Tris-HCl; Tris-d11; deuterated Tris; spelling/case variants	34 / 34	$pK_a = 8.06$	$\text{TrisH}^+(+1) \rightleftharpoons \text{Tris}(0)$	Not included as a molecule; calculated contribution added to the target ionic strength.	(Goldberg et al. 2002)
Chelator						
EDTA	EDTA; NaEDTA	15 / 15	$pK_a = 2.00, 2.69, 6.13, 10.37$	$\text{H}_4\text{Y}(0) \rightleftharpoons \text{H}_3\text{Y}^-(-1) \rightleftharpoons \text{H}_2\text{Y}^{2-}(-2) \rightleftharpoons \text{HY}^{3-}(-3) \rightleftharpoons \text{Y}^{4-}(-4)$	Not included as a molecule; calculated contribution added to the target ionic strength and represented with Na^+ and Cl^- .	(Smith et al. 2004; Hasegawa et al. 2013)

Table 1: BMRB sample-component names, grouped by the chemical entry used for the 173 retained systems (Ulrich et al. 2008). Counts are component rows and systems. Linked PDB records supplied clarification where needed (Berman et al. 2000); sources in the final column supplied the adopted values and charge states. The MES row distinguishes the executed charge states from the cited states.

Chemical entry	Recorded names	Rows / systems	Value used	State or dissociation used	Treatment in the prepared system	Source of value
Salts						
NaCl	NaCl; sodium chloride	98 / 98	$I/C = 1$	$\text{NaCl} \rightarrow \text{Na}^+ + \text{Cl}^-$	Na^+ and Cl^- included explicitly.	(International Union of Pure and Applied Chemistry 2025)
KCl	KCl; potassium chloride	15 / 15	$I/C = 1$	$\text{KCl} \rightarrow \text{K}^+ + \text{Cl}^-$	K^+ and Cl^- included explicitly.	(International Union of Pure and Applied Chemistry 2025)
CaCl_2	CaCl_2	18 / 18	$I/C = 3$	$\text{CaCl}_2 \rightarrow \text{Ca}^{2+} + 2\text{Cl}^-$	Ca^{2+} and Cl^- included explicitly.	(International Union of Pure and Applied Chemistry 2025)
Na_2SO_4	Na_2SO_4	1 / 1	$I/C = 3$	$\text{Na}_2\text{SO}_4 \rightarrow 2\text{Na}^+ + \text{SO}_4^{2-}$	Salt not included; its calculated contribution was represented with Na^+ and Cl^- .	(International Union of Pure and Applied Chemistry 2025)
Na_2HPO_4	Na_2HPO_4	1 / 1	$I/C = 3$	$\text{Na}_2\text{HPO}_4 \rightarrow 2\text{Na}^+ + \text{HPO}_4^{2-}$	Salt not included; its calculated contribution was represented with Na^+ and Cl^- .	(International Union of Pure and Applied Chemistry 2025)
KH_2PO_4	KH_2PO_4 ; KiPO_4	4 / 4	$I/C = 1$	$\text{KH}_2\text{PO}_4 \rightarrow \text{K}^+ + \text{H}_2\text{PO}_4^-$	Salt not included; its calculated contribution was represented with Na^+ and Cl^- .	(International Union of Pure and Applied Chemistry 2025)
$(\text{NH}_4)_2\text{SO}_4$	ammonium sulfate	1 / 1	$I/C = 3$	$(\text{NH}_4)_2\text{SO}_4 \rightarrow 2\text{NH}_4^+ + \text{SO}_4^{2-}$	NH_4^+ included explicitly; sulfate not included.	(International Union of Pure and Applied Chemistry 2025)
$\text{NH}_4\text{CH}_3\text{COO}$	NH_4OAc	5 / 5	$I/C = 1$	$\text{NH}_4\text{CH}_3\text{COO} \rightarrow \text{NH}_4^+ + \text{CH}_3\text{COO}^-$	Salt not included; its calculated contribution was represented with Na^+ and Cl^- .	(International Union of Pure and Applied Chemistry 2025)
NaCH_3COO	sodium acetate D3	2 / 2	$I/C = 1$	$\text{NaCH}_3\text{COO} \rightarrow \text{Na}^+ + \text{CH}_3\text{COO}^-$	Salt not included; its calculated contribution was represented with Na^+ and Cl^- .	(International Union of Pure and Applied Chemistry 2025)
CaCl_2 [interpreted]	CaCl	1 / 1	$I/C = 3$	Recorded as CaCl; treated as $\text{CaCl}_2 \rightarrow \text{Ca}^{2+} + 2\text{Cl}^-$.	Ca^{2+} and Cl^- included explicitly.	(International Union of Pure and Applied Chemistry 2025); preparation decision
Reducing agents						
DTT	DTT; d-DTT; dithiothreitol	61 / 60	Ionic-strength contribution set to zero	Recorded reducing agent	Not included as a molecule.	Preparation rule; identity: (Cleland 1964; Whitesides et al. 1977)
DTE	DTE	1 / 1	Ionic-strength contribution set to zero	Recorded reducing agent	Not included as a molecule.	Preparation rule; identity: (Cleland 1964; Whitesides et al. 1977)
β -mercaptoethanol	B-mercaptoethanol; 2-mercaptoethanol	2 / 2	Ionic-strength contribution set to zero	Recorded reducing agent	Not included as a molecule.	Preparation rule; identity: (Whitesides et al. 1977)
Preservative						
Sodium azide	NaN_3	1 / 1	$M = 65.0099 \text{ g mol}^{-1}$; $0.05\% \text{ w/v} = 7.69 \text{ mM}$; $I/C = 1$	$\text{NaN}_3 \rightarrow \text{Na}^+ + \text{N}_3^-$	Not included; contribution represented with Na^+ and Cl^- .	Molar mass: (National Institute of Standards and Technology n.d.); w/v interpretation: preparation decision

Table 2: BMRB sample-component names, grouped by the chemical entry used in preparation. Counts are component rows and systems; the fourth column gives the adopted value or rule. Explicit ions were represented using the GROMACS implementation of amber14SB (Abraham et al. 2015; Maier et al. 2015).

Sources

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